MIL-T-46603A(MU)
26 September 1969
SUPERSEDING
MIL-T-46603 (ORD)
27 November 1961

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M19, ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated: M19, Assembly, loaded with Cartridge, Impulse, M119, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-H-25579	- Hose Assembly, Tetrafluoroethylene, High Temperature, Power Plant, Aircraft

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MS-28741	- Hose Assembly-Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10520987	- Thruster, Cartridge Actuated, M19, Assembly
C8836128	- Container, Ammunition, Fiber M445 for Thruster, Cartridge Actuated, M19 or M13 with Cartridge M119

FSC 1377

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DRAWINGS (Cont'd)

C8836129	- Box, Packing, Ammunition for Thruster, Cartridge Actuated, M19 or M13 with Cartridge M119 in Fiber Container M445
IEL 10520987	- Inspection Equipment List for Thruster, Cartridge Actuated, M19, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Table I

Components	A Run-on (minimum) (inch pounds)	B Seating or Bottoming (including A) (inch pounds)	C Breakaway (inch pounds) (minimum)
Body and Base Cap	30	300 ± 20	100

3.3 Residual magnetism.- The thruster shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 The thruster shall not fail to fire.

3.4.2 The thruster shall function without mechanical failure (see 6.2).

3.4.3 No load.- The thruster fired with no load on the piston shall complete its stroke of 2.76 inches, minimum, and the piston shall not separate from the body of the thruster (see 6.3).

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3.4.4 Load.- The thruster fired under load shall perform as follows:

3.4.4.1 Restraining force.- With the piston connected to a 75 pound weight, the piston shall push the weight along a horizontal path of travel and the moving piston and weight shall overcome a restraining force of from 25 pounds at zero stroke to 500 pounds at 2.40 inches of stroke.

3.4.4.2 Thrust.- The thrust developed shall not exceed 2500 pounds after an elapsed time of 16 milliseconds.

3.4.4.3 By-pass pressure.- Furnish a by-pass pressure of 1000 pounds per square inch (psi) minimum, at the end of a 13 foot, 6 inch length of aircraft hose, size 4, MIL-H-25579.

3.4.4.4 Stroke.- Complete a piston stroke ranging from 2.76 inches, minimum, to 2.86 inches, maximum.

3.4.4.5 Time.- Time between the firing of the cartridge and completion of the stroke shall not exceed 80 milliseconds.

3.4.4.6 Ignition delay.- Ignition delay shall not exceed 25 milliseconds. Ignition delay is defined as the time lapse between energizing of the primer and the first continuous pressure rise in the body of the thruster.

3.4.5 Temperature.- The thruster shall comply with all the ballistic requirements throughout the temperature range of minus 65 degrees to plus 200 degrees F.

3.4.6 Initiation.- The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size no. 4, MS28741, @ $70^{\circ} \pm 5^{\circ}$F.

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defects.

3.5.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.5.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots, lot size and presentation of lots shall be in accordance with the "Submission of Product", provision of MIL-STD-105.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10000	200	2	3	3	4

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4.2.2.1 Cartridge, Impulse, M119, Dwg. D8595513, covering a detail of Dwg. F10520987.

	Defect	Method of Inspection
CRITICAL		
1.	Mixed cartridges	Visual
2.	Metal defective	Visual
3.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Foreign matter	Visual
MINOR		
201.	Case scratched, wrinkled, bulged or dented	Visual

4.2.2.2 Assembly, Dwg. F10520987, less Shipping Caps, Dwg. A8593299 and Base Cap Assembly Dwg. C10520975.

CRITICAL		
1.	Shear pin not engaged in firing pin	Gage
2.	Metal defective	Visual
MAJOR		
101.	Total length from center of mounting lug to face of piston fastener	Gage
102.	Total length	Gage
103.	Thread damaged	Visual
104.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.3 Assembly, Dwg. F10520987.

CRITICAL		
1.	Any part missing, as detected by x-ray	X-ray
2.	Improper assembly	X-ray

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
CRITICAL	
3. Metal defective	Visual
4. Foreign matter in gas ports	Visual
MAJOR	
101. Marking missing, incorrect or illegible	Visual
MINOR	
201. Total length, max.	Gage
202. Set screws not filled with barrier material	Visual
203. Protective finish defective	Visual
204. Shipping cap or plug missing	Visual
205. Location of labels, incorrect	Visual

4.2.2.4 Fiber Container, Dwg. C8836128.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

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4.2.2.5 Box, Packing, Dwg. C8836129.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual & Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual & Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking misleading or unidentifiable	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- Each base cap and body assembly shall be subjected to a breakaway torque application. Failure of any thruster to comply with the requirements of 3.2 shall be cause for rejection of the thruster.

4.2.3.2 Residual magnetism.- Each thruster shall be tested for excessive residual magnetism. Failure of any thruster to comply with the requirement of 3.3 shall be cause for rejection of the thruster.

4.2.3.3 Functional tests.- The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics.- Twenty-one units selected at random shall be tested to determine compliance with the requirements of 3.4.1 thru 3.4.6. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

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Table III

Sample Size	Method	Temperature
6	Load (see 3.4.4)	-65° to -70°F
6	Load (see 3.4.4)	+65° to +75°F
6	Load (see 3.4.4)	+200° to +205°F
1	No Load (see 3.4.3)	-65° to -70°F
1	No Load (see 3.4.3)	+65° to +75°F
1	No Load (see 3.4.3)	+200° to +205°F

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10520987.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, capable of measuring the torque requirements specified in column C, Table I in 3.2 shall be used with an appropriate adapter to determine the adequacy of the joint. The thruster body shall be secured and the breakaway torque shall be applied to the base cap.

4.3.2 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation.

4.3.3 Functional tests.

4.3.3.1 Ballistics.- The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures.

4.3.3.1.1 Temperature conditioning time.- A simulated thruster and cartridge with a temperature sensitive element, placed in a dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A.- Packing and marking shall be in accordance with Dwgs. C8836128 and C8836129.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 No load.- Permanent yielding of the mounting lugs and piston shall not be cause for rejection when thrusters are tested with no load on the piston.

6.4 Load.- The expected performance of the thruster under conditions simulating those existing in aircraft is obtained when thruster satisfactorily meets conditions of the load firing.

Custodian:

Army - MU

Preparing activity:

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Project No. 1377-A372

*U.S. GOVERNMENT PRINTING OFFICE: 1971-714-161/4027

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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